

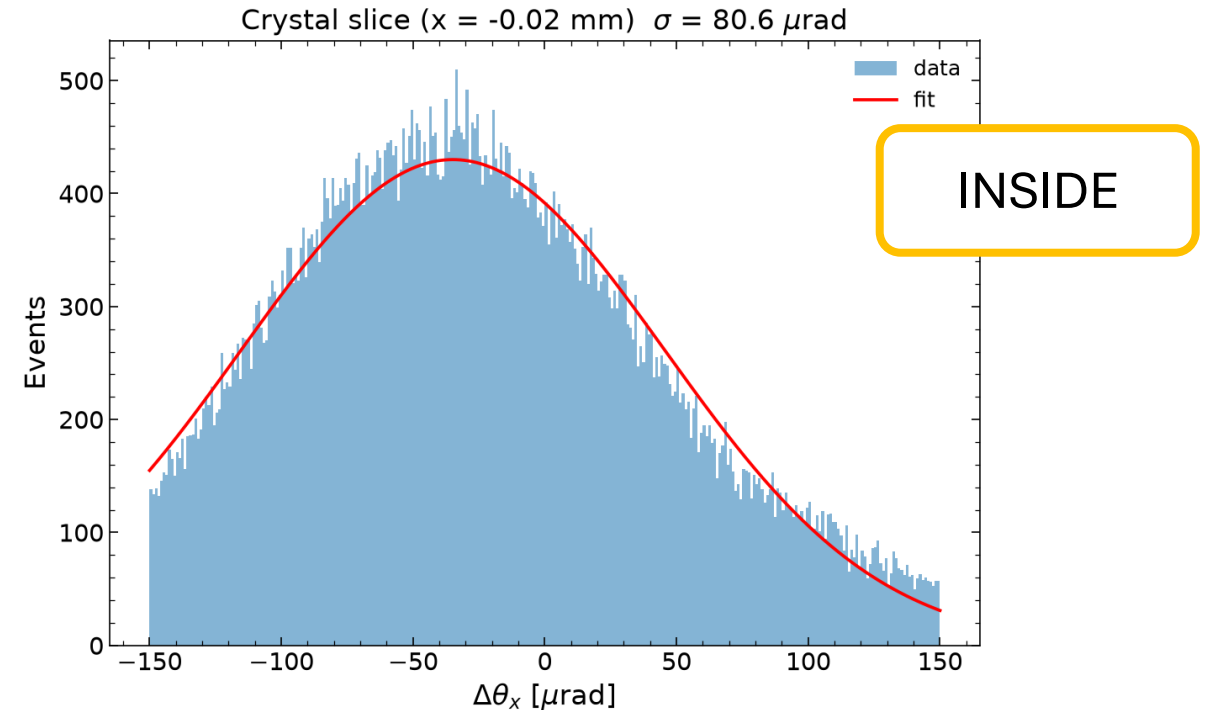
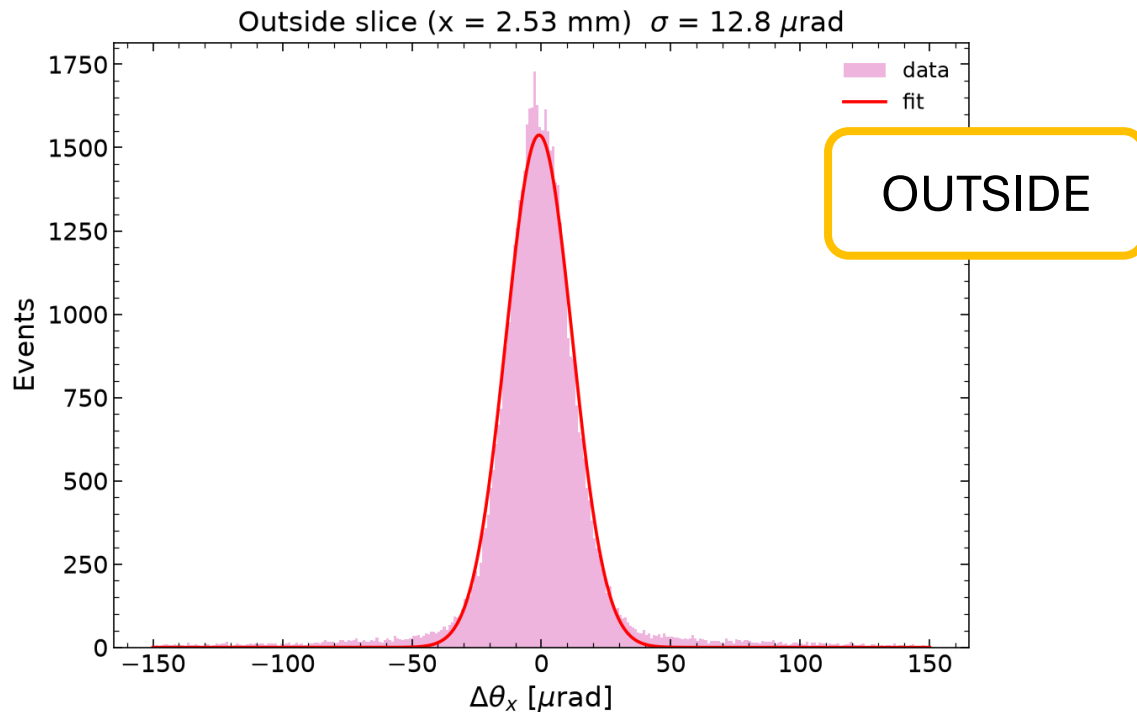
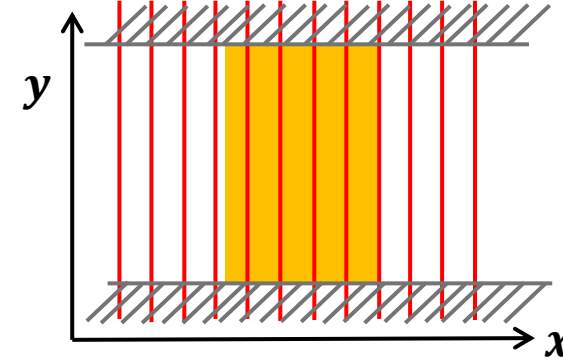
SEPTEMBER 2026 UPDATE
ANNA SOFIA MORO

WHAT'S NEW?

1. Multiple Coulomb Scattering
 - Analysis made for both x and y
 - Clamp material evaluation
2. Analysis of the beam distribution of real data
 - Check the distributions
 - Check the correlation/covariance
3. Simulation
 - Reproduce the real data beam features
 - Channeling efficiency analysis
4. Thesis
 - Possible order of topics in the table of contents

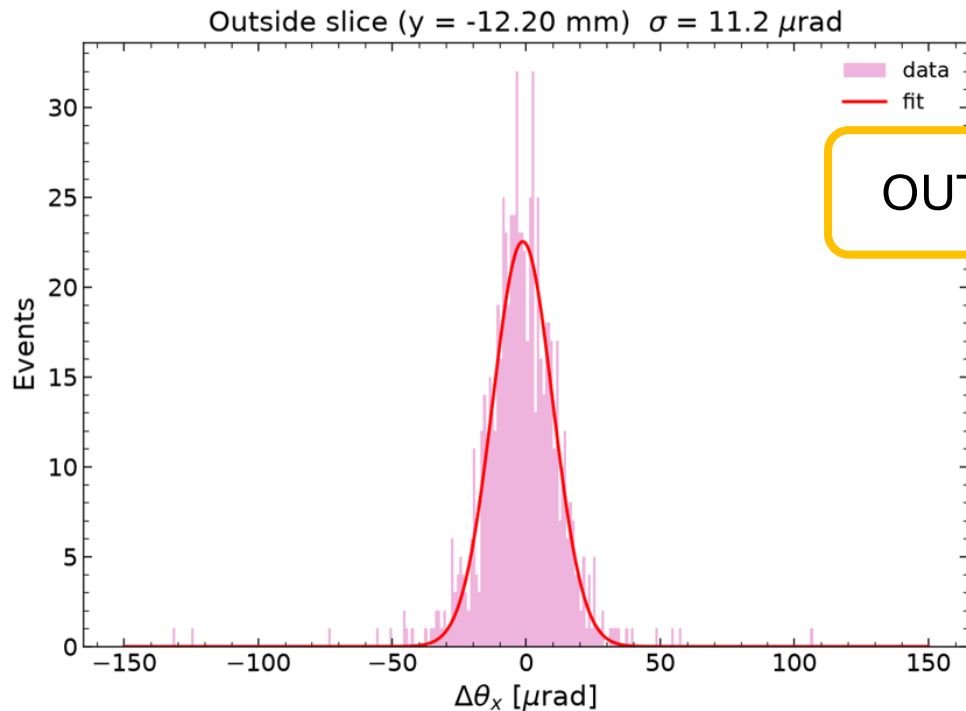
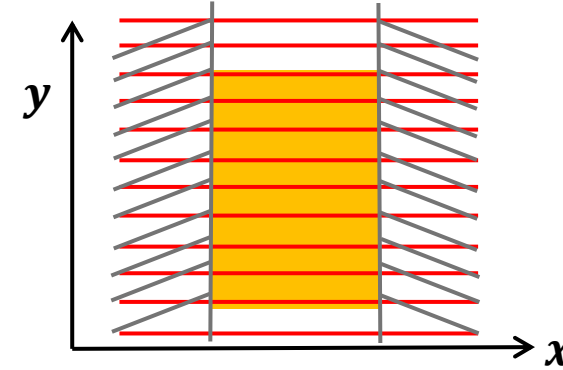
FIND THE CRYSTAL FOOTPRINT THROUGH SCATTERING

- Divided the crystal into slices along x or y
- Width of the angular distribution σ for each slice

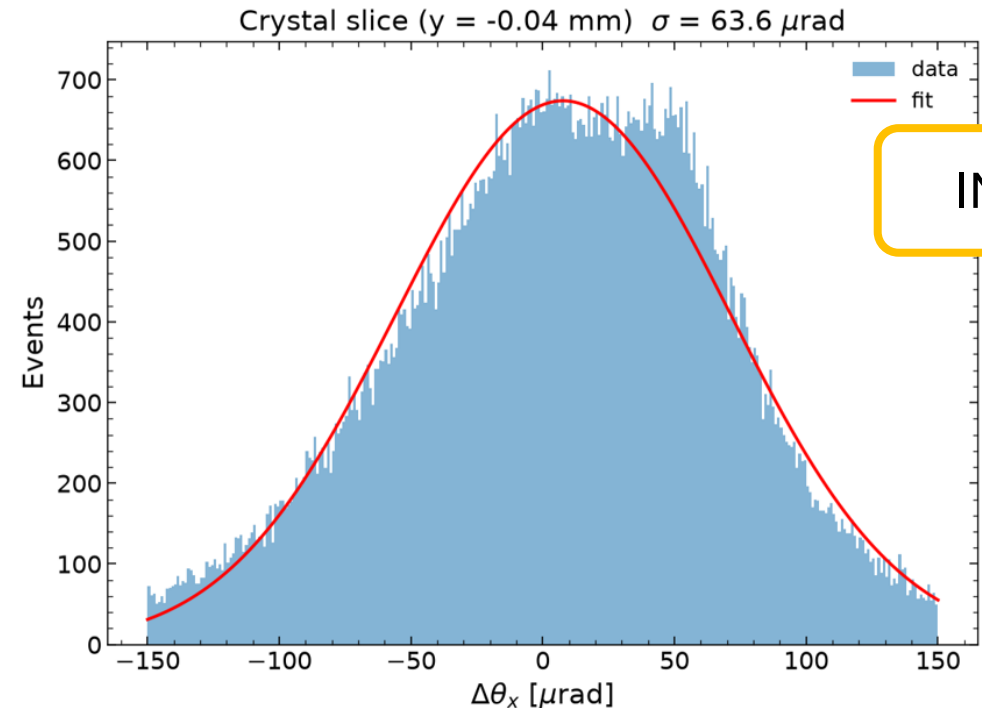


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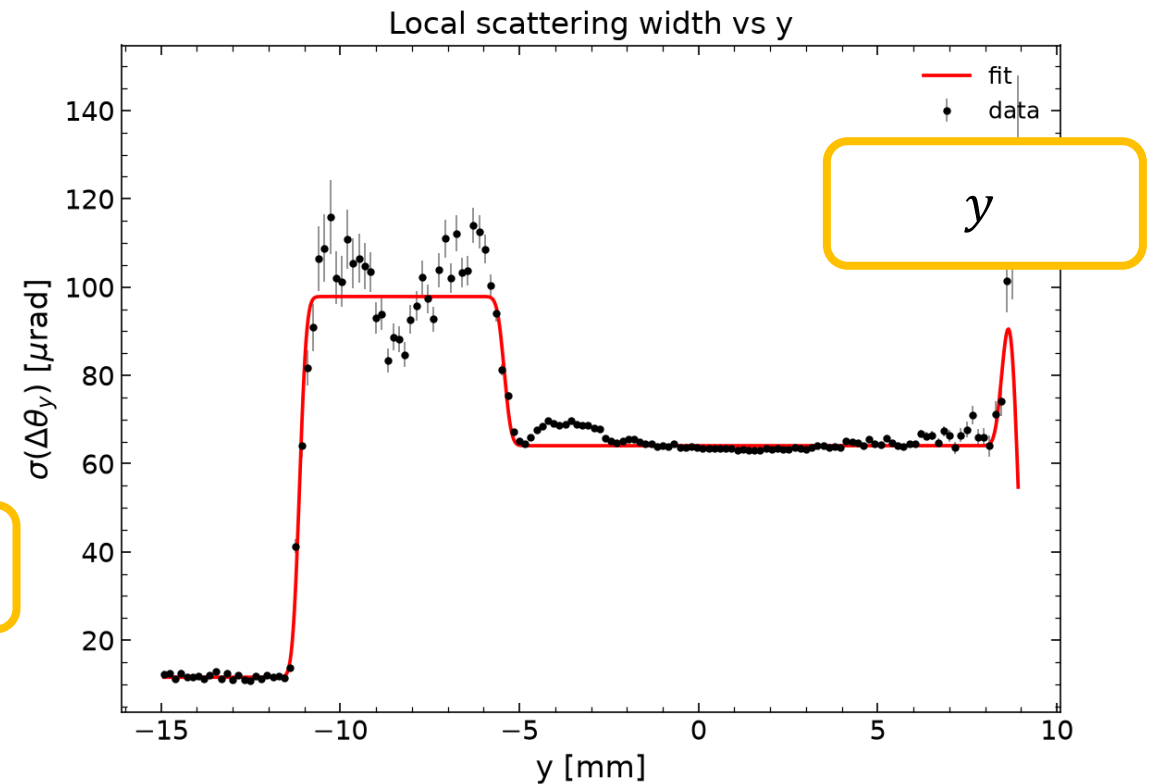
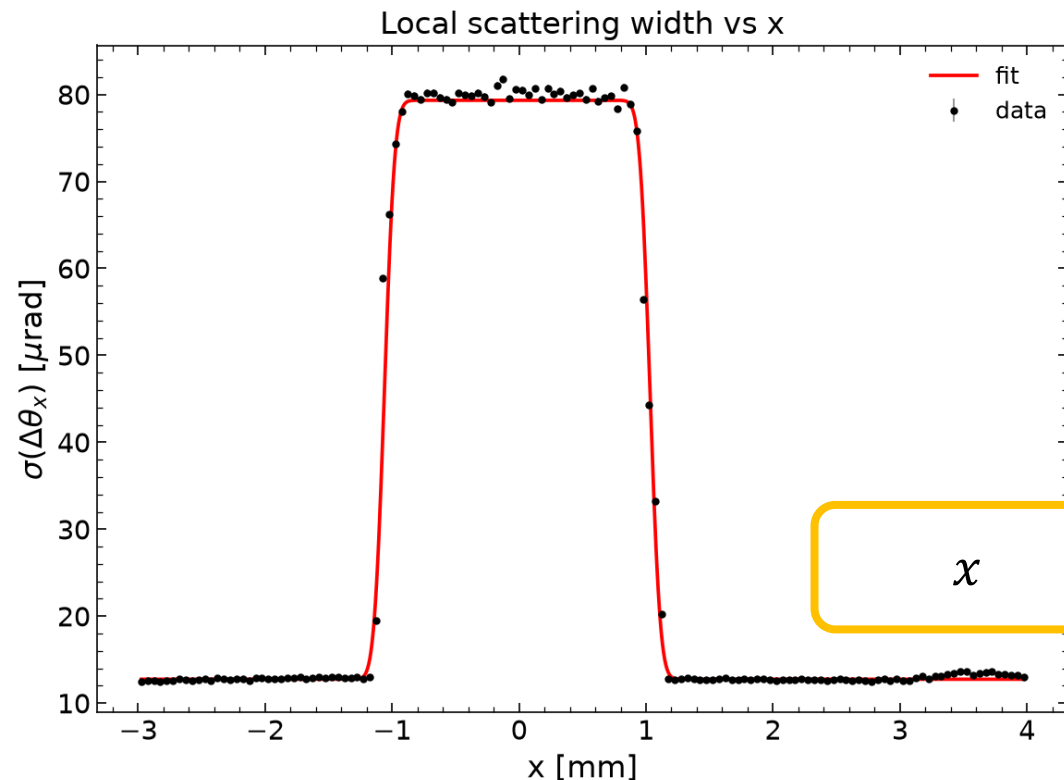
OUTSIDE



INSIDE

FIND THE CRYSTAL FOOTPRINT THROUGH SCATTERING

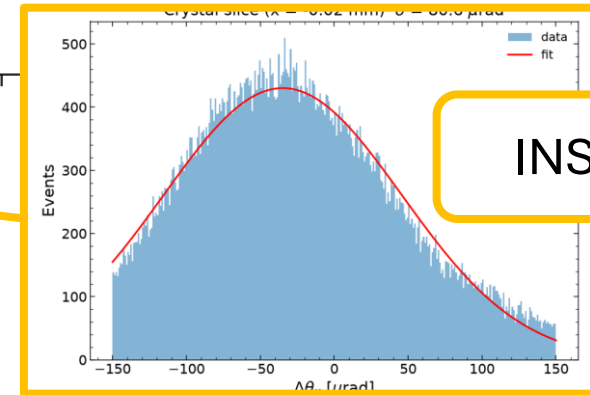
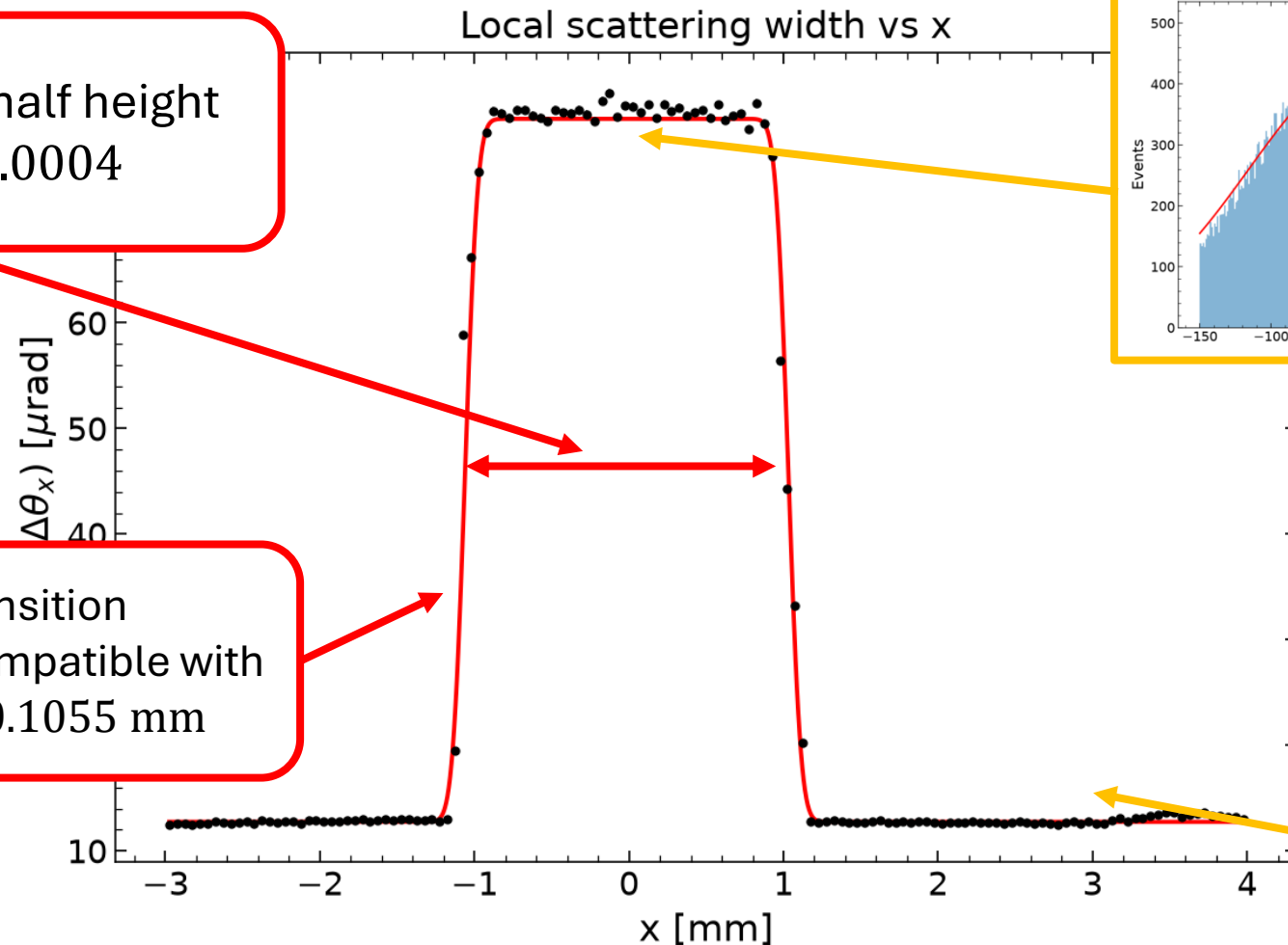
- The extracted gaussian widths σ are plotted against the impact position
- The transition between the outside and inside regions is fitted using a Error function (Erf)



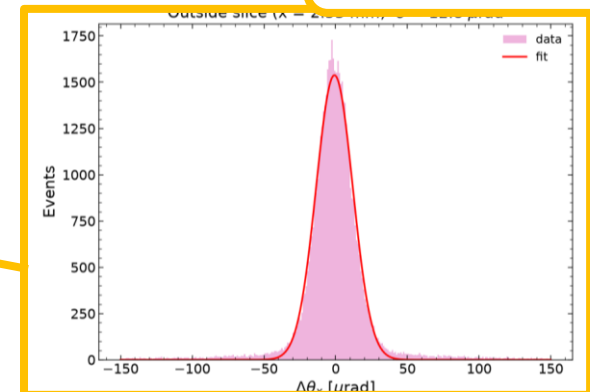
FIND THE CRYSTAL FOOTPRINT THROUGH SCATTERING

The window @ half height
 2.0911 ± 0.0004

The width transition
 $0.0901(3)$ mm is compatible with
position error ~ 0.1055 mm



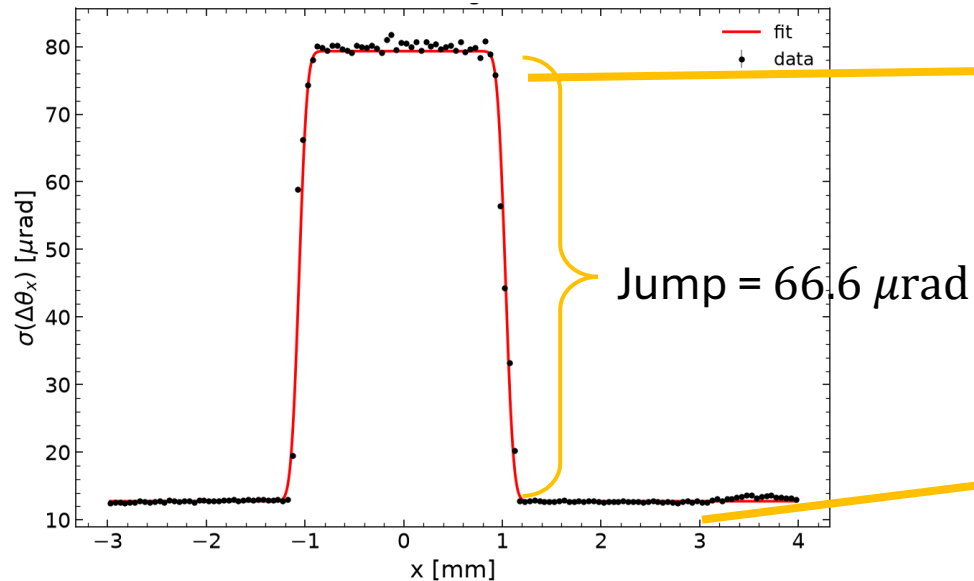
INSIDE



OUTSIDE

FIND THE CRYSTAL FOOTPRINT THROUGH SCATTERING

$$\langle \theta_{MCS} \rangle = \frac{13.6 \text{ MeV}}{\beta c p} z \sqrt{\frac{x}{X_0}} \left[1 + 0.038 \ln \left(\frac{x z^2}{X_0 \beta^2} \right) \right] = 66.5 \mu\text{rad}$$



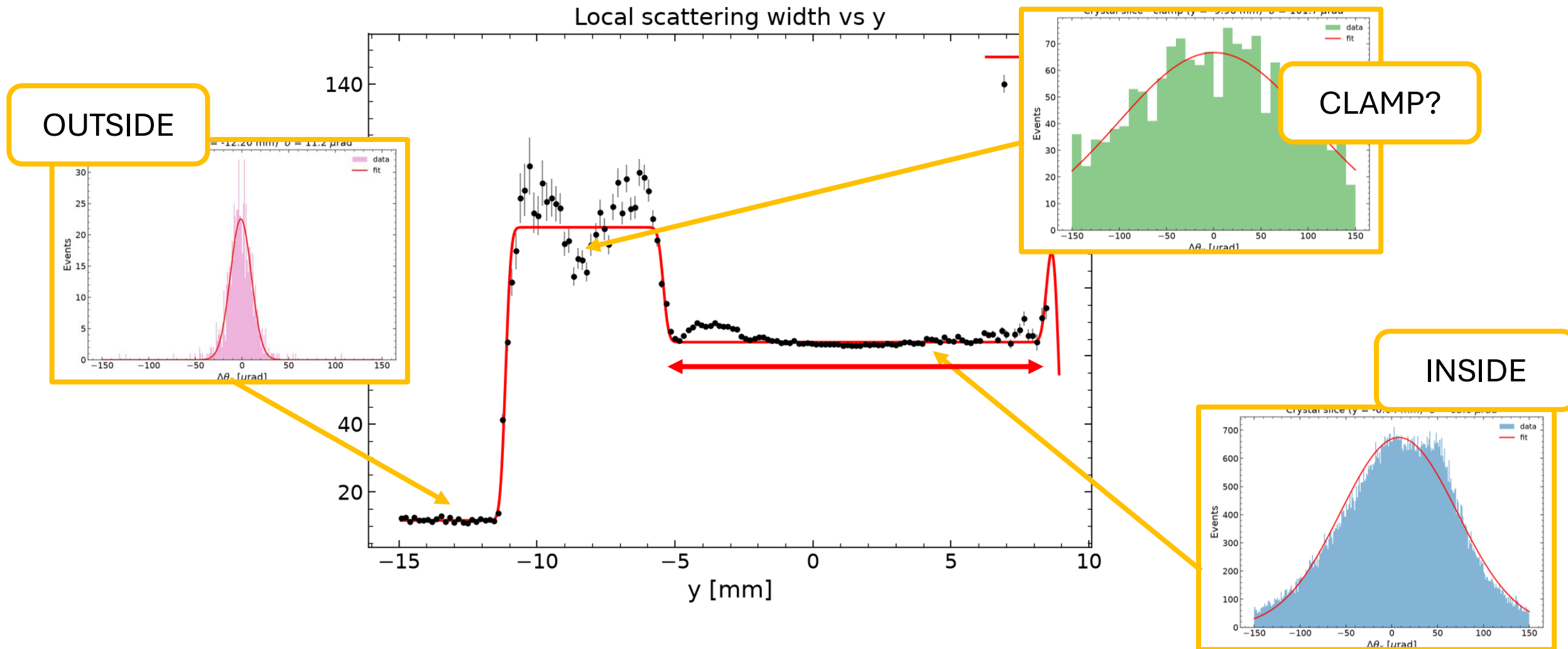
$$\sigma_{meas} = \sqrt{\sigma_{mcs}^2 + \sigma_{track}^2} \sim 79.5 \mu\text{rad}$$

$$\rightarrow \sigma_{mcs} = 78.30(6) \mu\text{rad} \neq \sigma_{mcs}^{th}$$

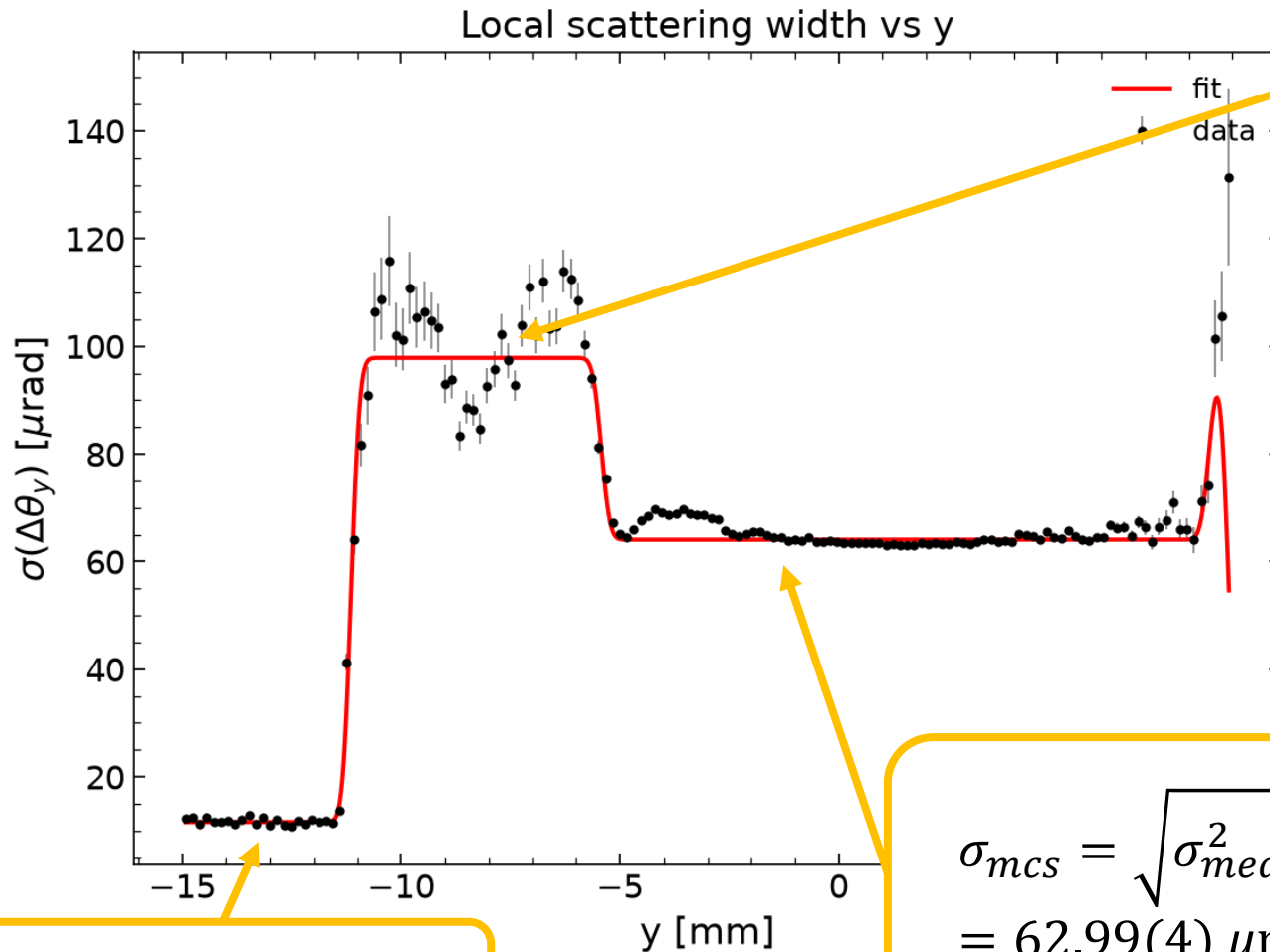
not zero, but $\sim 12.75(1) \mu\text{rad}$
which corresponds to σ_{track}

$$\sigma_{track} = \sqrt{\sigma_{track,in}^2 + \sigma_{track,out}^2} = 12.6 \mu\text{rad}$$

FIND THE CRYSTAL FOOTPRINT THROUGH SCATTERING



FIND THE CRYSTAL FOOTPRINT THROUGH SCATTERING



CLAMP: which material/metal?
on average $\sim 99.2 \pm 0.3 \mu\text{rad}$

unknown: x, X_0

Between different metallic materials (stainless steel, aluminium, titanium, copper, brass):

→ The only similar is titanium

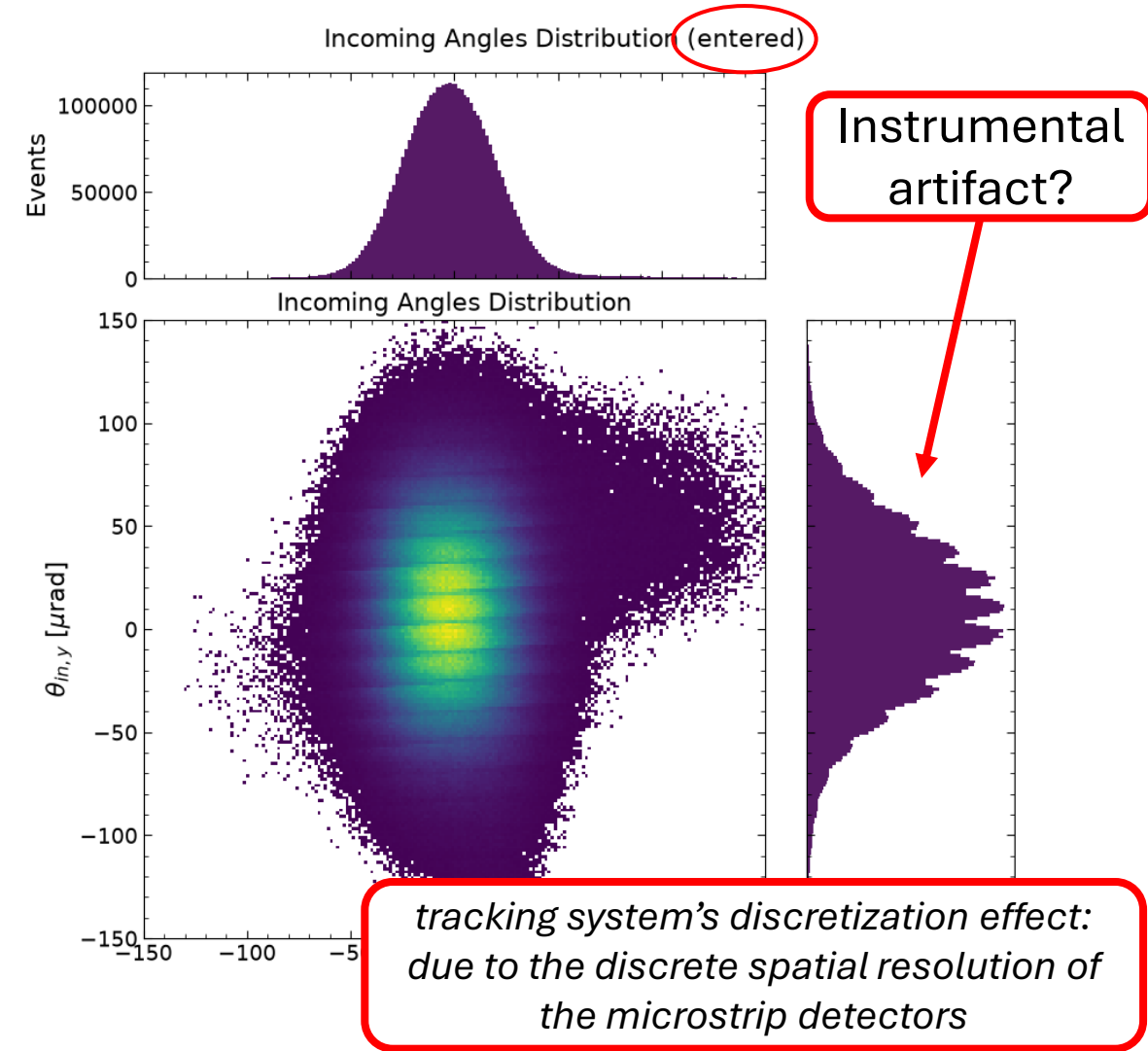
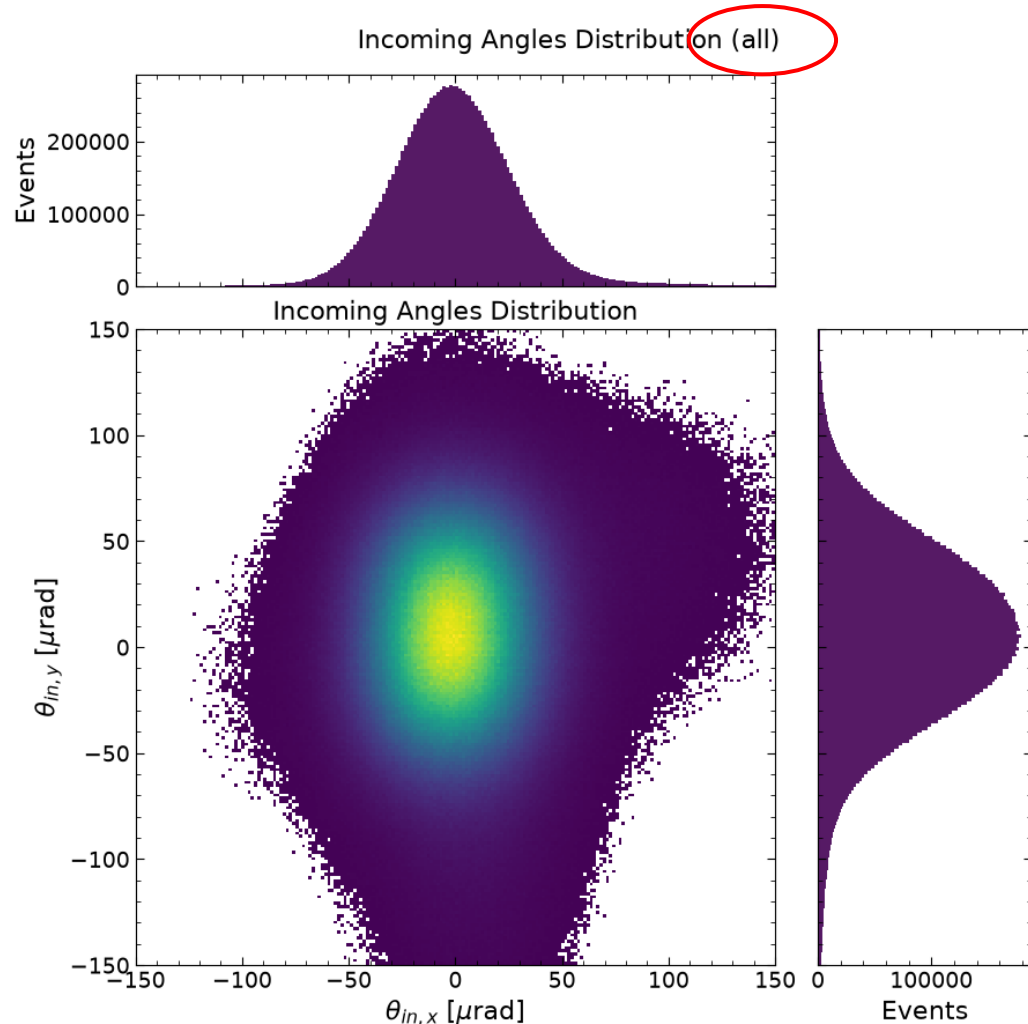
$\theta \sim 110 \mu\text{rad}$

(stainless steel too much scattering, aluminium not enough scattering)

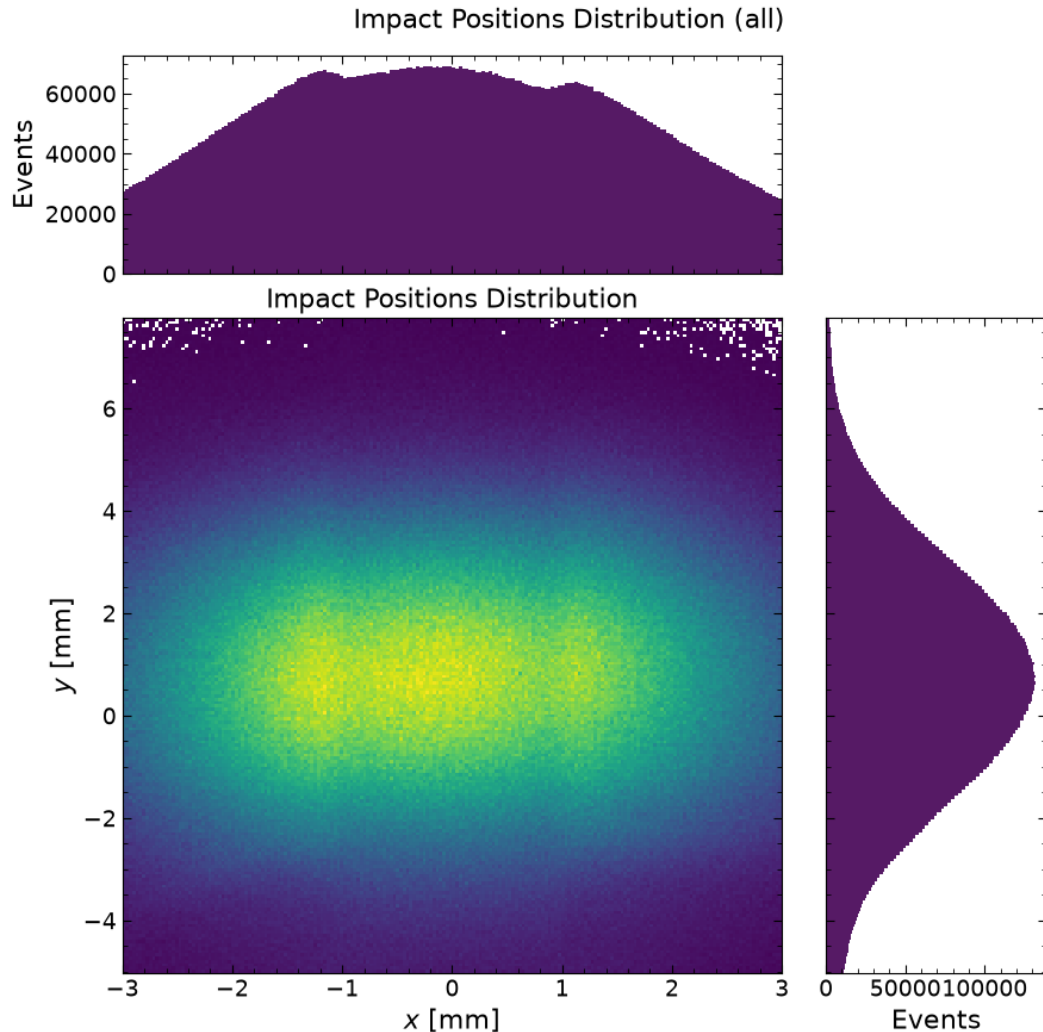
$$\sigma_{bsl} = 11.6(1) \mu\text{rad}$$

$$\sigma_{mcs} = \sqrt{\sigma_{meas}^2 - \sigma_{bsl}^2}$$
$$= 62.99(4) \mu\text{rad}$$

ANALYSIS OF THE BEAM DISTRIBUTION (REAL DATA)



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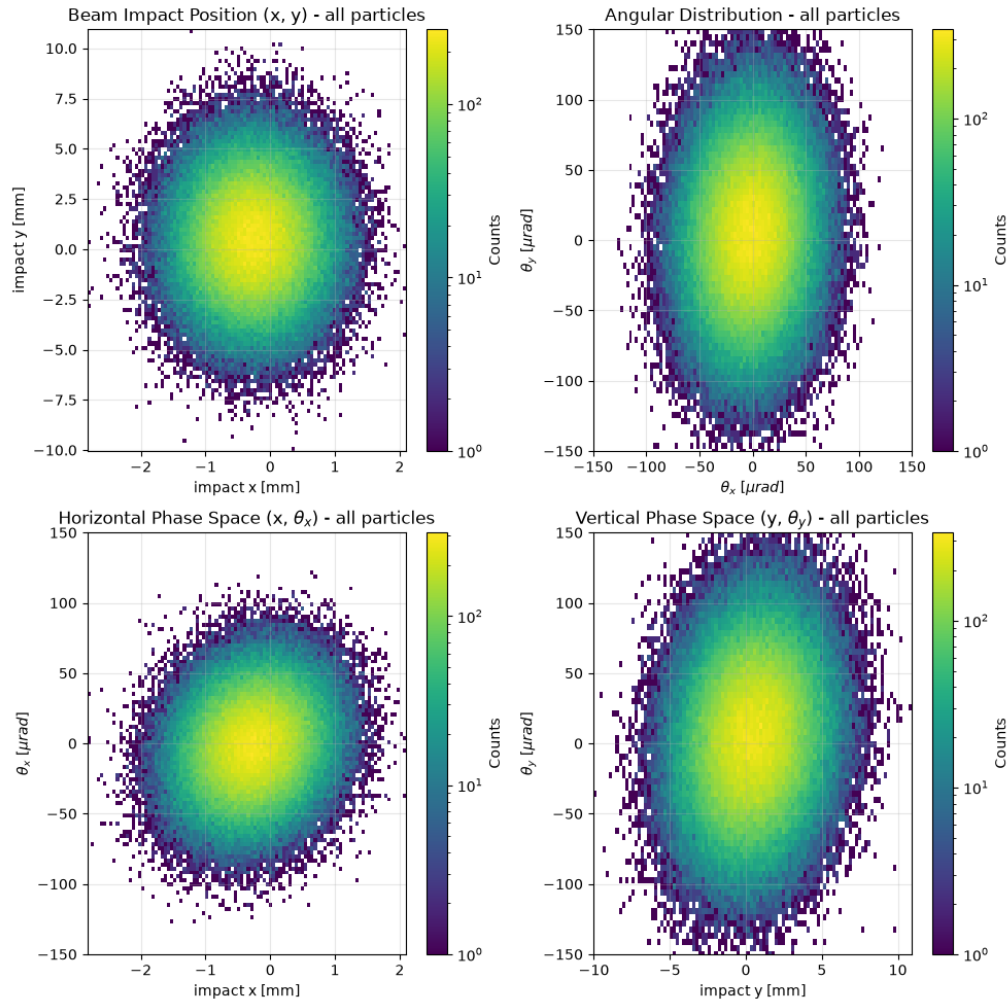
✓ Gaussian fits on the distributions of x, y, θ_x, θ_y

✓ Check the correlation:

Not negligible for $x - \theta_x$, $y - \theta_y$:

	x	y	θ_x	θ_y
x	1	0	0.2	0
y	0	1	0	0.2
θ_x	0.2	0	1	0
θ_y	0	0.2	0	1

SIMULATING THE SHORT CRYSTAL



Fitting the 1d marginal distributions and then sampling them independently would not be correct if variables are uncorrelated.

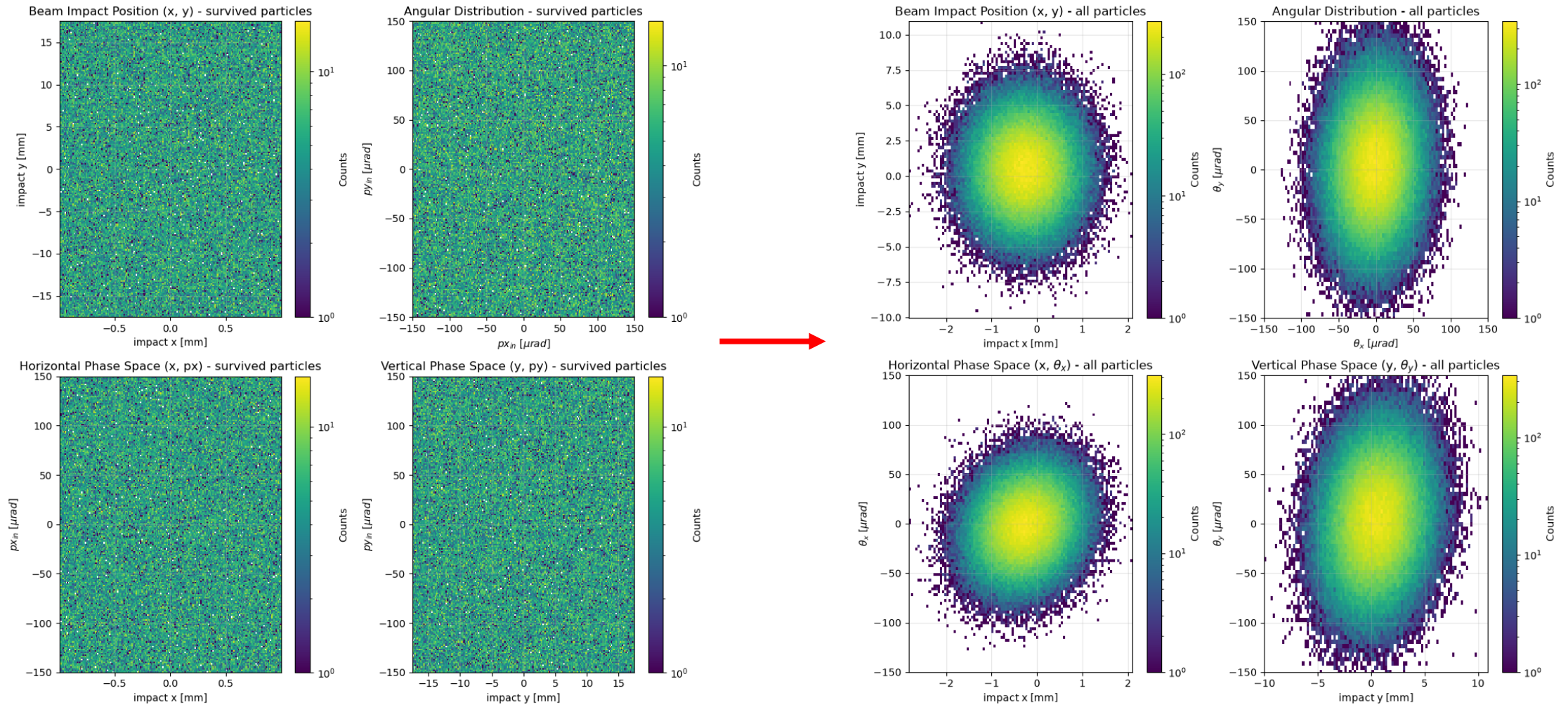
Solution:

Using a multivariate random normal distribution (thanks to the covariance matrix reproduce the correlation)

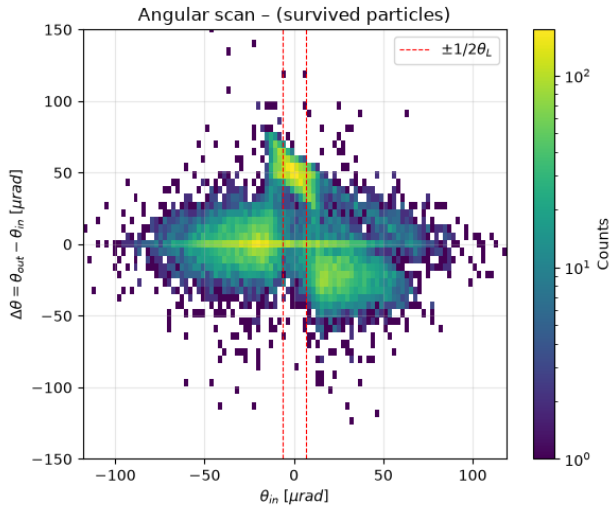
SIMULATING THE SHORT CRYSTAL

Before: normal distribution

Now: normal distribution

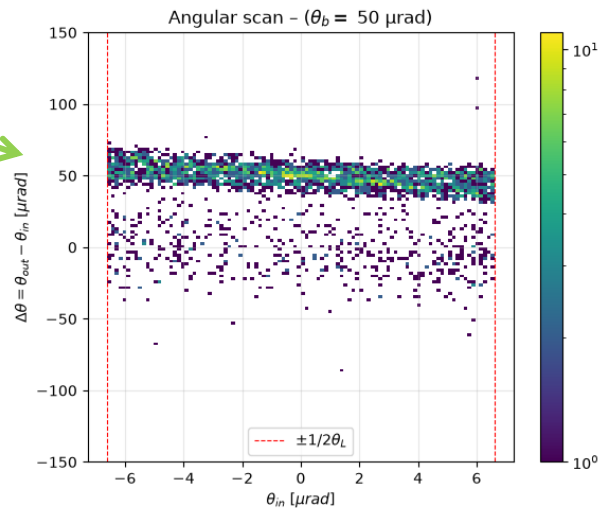


SIMULATING THE SHORT CRYSTAL



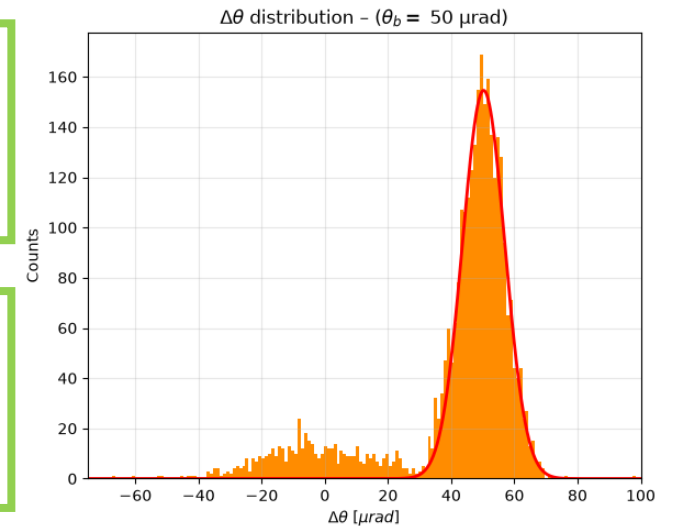
Selecting particles that entered in the crystal:
compatible with theoretical expectation $f = \text{erf}\left(\frac{1}{\sigma_x\sqrt{2}}\right) = 92\%$

Also performed the edges analysis: Multiple Coulomb scattering
and Channeled particles footprint

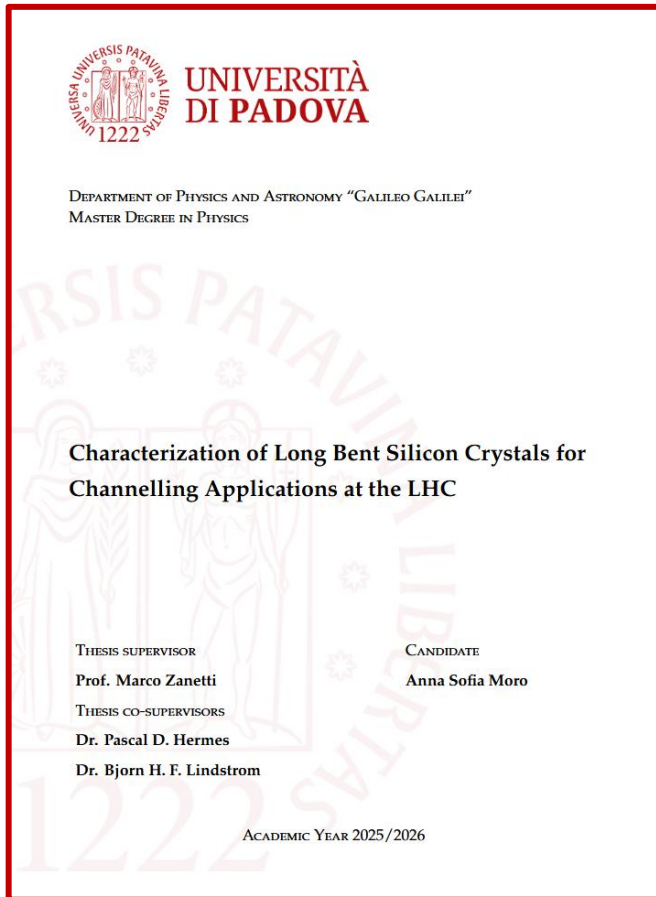


Selecting particles within $\pm\theta_L/2$

$$\epsilon = N_{ch}/N_{tot} \sim 75\%$$



THESIS: PART I



- Physics background
 - Dipole moment
 - Λ_C^+
- Crystal Channeling
 - In straight/bent crystals
 - Other processes (dechannelling, volume capture, ...)
- Experimental principle: measuring spin precession through bent crystals
- TWOCRIST experiment
- H8 testing